

Landing-Only Linear Search for an Escaping Target: Discovery and Capture with Known and Unknown Parameters

Abstract

We study online search on the real line by a unit-speed flying searcher for a target that starts at distance at least one from the origin and moves directly away from the origin at constant speed $v < 1$. Unlike the classical linear-search model, the searcher cannot observe the target while flying: detection is possible only at a landing, and only when the target lies between the origin and the landing point. We distinguish *discovery*, in which the process stops when the target is detected, from *capture*, in which the searcher must subsequently meet the target.

Four information regimes are considered. When the target's initial distance is unknown but its speed is known, we give matching upper and lower bounds for both discovery and capture. The optimal discovery ratio is

$$\frac{(\sqrt{5+4v}+3)(2v+\sqrt{5+4v}+1)^2}{4(1-v)^2(\sqrt{5+4v}-1)},$$

whereas the optimal capture ratio is

$$\frac{(2v+1)(v+5)^2}{2(1-v)^2(1+v)}.$$

When the initial distance magnitude is known but its sign and the speed are unknown, we introduce an implementable doubly-exponential landing strategy. Both discovery and capture admit speed-dependent upper bounds with asymptotic exponent $23/4$:

$$\text{CR}(v) = O((1-v)^{-23/4}), \quad v \rightarrow 1^-.$$

The exponent follows by optimizing excursions $a_i = A^{b^i}$ over $b > 1$, with the optimum $b = 3/2$. We conclude with a sequence-theoretic conjecture suggesting that $23/4$ is also the universal lower-bound exponent in the unknown-speed, known-distance regime.

1 Introduction

Online search asks how efficiently a mobile agent can locate an object whose relevant parameters are not known in advance. The classical one-dimensional problem, often called linear search or the cow-path problem, places a stationary target at an unknown point of the line and compares an online zig-zag trajectory with a clairvoyant searcher. Its characteristic feature is geometric expansion: the searcher alternates directions while increasing the distance explored on successive excursions.

The present paper studies a mobile analogue with two additional constraints. First, the target is *escaping*: if it starts at $d > 0$ it moves to the right, and if it starts at $d < 0$ it moves to the left, always with a fixed speed $v < 1$. Second, the searcher is a flying agent with *landing-only visibility*. It receives no information while in flight. A target can be discovered only at a landing, and then only if the target lies on the segment joining the origin to the landing point. This changes the usual interception geometry substantially: overshooting the target while flying does not reveal it.

We distinguish two objectives. In the *discovery* problem the cost is the time of the first successful landing. In the *capture* problem the searcher must, after discovery, physically meet the target. Once a landing discovers the target, the target's speed can be inferred from its observed position, the known elapsed time, and the initial distance when that distance is known; alternatively, for the cost calculation it is enough to note that an inward-moving searcher and outward-moving target close at relative speed $1 + v$.

The amount of prior information has a decisive effect. We consider the following four regimes:

- (i) discovery, initial distance unknown, speed known;
- (ii) capture, initial distance unknown, speed known;
- (iii) discovery, initial distance magnitude known but sign and speed unknown;
- (iv) capture, initial distance magnitude known but sign and speed unknown.

For the known-speed cases we obtain exact optimal competitive ratios. For the unknown-speed cases no uniform constant ratio is expected; instead we measure performance as a function of the actual speed v and obtain doubly-exponential strategies with exponent $23/4$ as $v \rightarrow 1^-$. The same exponent appears for discovery and capture because, at the relevant scale, both costs are dominated by the final doubly-exponential excursion.

1.1 Related work

The classical linear-search problem and its star-search generalizations form a basic family of online-search problems. The optimal deterministic competitive ratio on the line is 9, achieved by alternating geometric excursions with doubling; this result is commonly associated with the classical work of Beck and Newman and has been repeatedly rediscovered and generalized. Demaine, Fekete, and Gal studied linear and star search in the presence of a fixed turn cost and showed how infinite linear-programming and recurrence arguments can yield tight competitive bounds [1]. Bose and De Carufel subsequently developed a broader framework allowing different outward and inward travel costs [2].

Escaping-target search replaces the stationary object by a target moving away from the origin. Recent work of Coleman, Ivanov, Kranakis, Krizanc, and Morales-Ponce studies the power of knowing the target speed and/or its initial distance in the standard continuously observable interception model, including the cases of unknown speed and known or unknown initial distance [3]. The information regimes considered there motivate the four-way classification used here. Our model differs in a crucial sensing assumption: passing a target in flight does not reveal it. Detection occurs only at designated landing points, so each excursion has an *effective reach* that depends both on its landing distance and on all time spent in previous excursions.

The landing-only restriction also has a natural connection with robotic search in which sensing is available only at checkpoints or after costly observations. Our purpose, however, is purely competitive and geometric: acceleration, sensing noise, and physical landing duration are ignored, and the searcher moves at unit speed.

1.2 Contributions and organization

Section 2 gives the common model and benchmark. Sections 3 and 4 give tight upper and lower bounds when v is known and the initial distance is unknown. Sections 5 and 6 treat the new unknown-speed setting when the distance magnitude is known. Section 8 records the conjectured matching lower-bound exponent for the latter setting.

2 Model and notation

A searcher starts at the origin of $\mathbb{R}$ and moves with maximum speed 1. A target starts at

$$d = \sigma D, \quad \sigma \in \{-1, +1\}, \quad D \geq 1,$$

and moves directly away from the origin at a constant speed

$$0 \leq v < 1.$$

Thus its position at time t is

$$y(t) = \sigma(D + vt).$$

The searcher cannot detect the target while flying. At a landing x at time t , it discovers the target if x has the correct sign and

$$|x| \geq D + vt.$$

Equivalently, the target lies between the origin and the landing point.

A standard alternating strategy has landing points

$$x_i = (-1)^i a_i, \quad a_i > 0,$$

and cumulative excursion lengths

$$S_i := \sum_{j=0}^i a_j, \quad S_{-1} := 0.$$

Because the searcher travels from one side through the origin to the other, the i th landing time is

$$t_i = 2S_{i-1} + a_i = 2S_i - a_i. \quad (1)$$

The *effective initial-distance reach* of landing i against speed v is

$$B_i(v) := a_i - vt_i = (1 + v)a_i - 2vS_i = (1 - v)a_i - 2vS_{i-1}. \quad (2)$$

A target on the side of excursion i is discovered at that landing exactly when $D \leq B_i(v)$.

If the sign, distance, and speed are known offline, a unit-speed searcher meets the target at

$$T^*(D, v) = \frac{D}{1 - v}. \quad (3)$$

We use this same clairvoyant benchmark for discovery and capture; a clairvoyant searcher can land exactly at the meeting point.

For a fixed known v and unknown D we use the ordinary worst-case competitive ratio. When D is known but v is unknown, we use the speed-dependent ratio

$$\text{CR}_{\mathcal{A}}(v) := \sup_{\sigma \in \{-1, +1\}} \frac{T_{\mathcal{A}}(\sigma D, v)}{D/(1-v)}.$$

Scaling permits us to normalize $D = 1$ in the unknown-speed sections.

3 Discovery with unknown distance and known speed

Assume v is known, whereas both $D \geq 1$ and its sign are unknown. Consider geometric excursions

$$a_i = r^i, \quad r > 1.$$

Then

$$S_i = \frac{r^{i+1} - 1}{r - 1}.$$

The effective reach is positive asymptotically only if

$$r > \frac{1+v}{1-v}. \tag{4}$$

Indeed,

$$\frac{B_i(v)}{a_i} \longrightarrow \frac{(1-v)r - (1+v)}{r-1}.$$

Suppose a target on the side of round i is first discoverable there. The preceding visit to that side was round $i-2$, so the worst target satisfies $D \downarrow B_{i-2}(v)$. Discovery costs t_i . Hence

$$\frac{t_i}{T^*} = (1-v) \frac{t_i}{D},$$

and asymptotically

$$\text{CR}_{\text{disc}}(r, v) = (1-v) \frac{r^2(r+1)}{(1-v)r - (1+v)}. \tag{5}$$

Theorem 3.1 (Optimal known-speed discovery ratio). *For landing-only discovery with unknown initial distance and known speed v , the optimal geometric expansion is*

$$r_{\text{disc}}^*(v) = \frac{1 + 2v + \sqrt{5 + 4v}}{2(1 - v)}, \quad (6)$$

and the optimal competitive ratio is

$$\boxed{\text{CR}_{\text{disc}}^*(v) = \frac{(\sqrt{5 + 4v} + 3)(2v + \sqrt{5 + 4v} + 1)^2}{4(1 - v)^2(\sqrt{5 + 4v} - 1)}}. \quad (7)$$

Moreover, this ratio is a lower bound for arbitrary deterministic alternating record strategies.

Upper bound. Ignoring the positive factor $1 - v$, minimize

$$f(r) = \frac{r^2(r + 1)}{(1 - v)r - (1 + v)}.$$

Differentiation gives

$$f'(r) = 0 \iff (1 - v)r^2 - (1 + 2v)r - (1 + v) = 0.$$

The feasible positive root is (6); substitution into (5) gives (7). $\square$

For completeness, the matching lower bound can be phrased as a sequence lemma. For an arbitrary useful alternating record sequence, $a_i = S_i - S_{i-1}$ and

$$t_i = S_i + S_{i-1}, \quad B_i(v) = (1 - v)S_i - (1 + v)S_{i-1}.$$

An adversary placing the target just beyond $B_{i-2}(v)$ gives

$$\text{CR} \geq (1 - v) \sup_i \frac{S_i + S_{i-1}}{(1 - v)S_{i-2} - (1 + v)S_{i-3}}. \quad (8)$$

The sharp sequence constant is

$$K_{\text{disc}}^*(v) = \min_{r > (1+v)/(1-v)} \frac{r^2(r + 1)}{(1 - v)r - (1 + v)}. \quad (9)$$

The associated critical recurrence

$$S_i + S_{i-1} = K((1 - v)S_{i-2} - (1 + v)S_{i-3})$$

has characteristic equality

$$r^3 + r^2 = K((1-v)r - (1+v)),$$

so its critical value is exactly (9). This yields the lower bound matching Theorem 3.1. At $v = 0$,

$$r^* = \frac{1 + \sqrt{5}}{2}, \quad \text{CR}_{\text{disc}}^*(0) = \frac{11 + 5\sqrt{5}}{2}.$$

As $v \rightarrow 1^-$,

$$\text{CR}_{\text{disc}}^*(v) \sim \frac{27}{(1-v)^2}.$$

4 Capture with unknown distance and known speed

We retain geometric excursions $a_i = r^i$ and the feasibility condition (4). Suppose the target is first discovered at landing i . At that time the searcher is beyond the target. It reverses direction; the searcher and target approach each other at relative speed $1 + v$. The total capture time is

$$\begin{aligned} T_i(D, v) &= t_i + \frac{a_i - D - vt_i}{1 + v} \\ &= \frac{t_i + a_i - D}{1 + v} = \frac{2S_i - D}{1 + v}. \end{aligned} \quad (10)$$

Consequently

$$\frac{T_i(D, v)}{T^*(D, v)} = \frac{1-v}{1+v} \left(\frac{2S_i}{D} - 1 \right). \quad (11)$$

For a target just beyond $B_{i-2}(v)$, geometric excursions give

$$\text{CR}_{\text{cap}}(r, v) = \frac{1-v}{1+v} \left[\frac{2r^3}{(1-v)r - (1+v)} - 1 \right]. \quad (12)$$

Theorem 4.1 (Optimal known-speed capture ratio). *The optimal geometric expansion for capture is*

$$r_{\text{cap}}^*(v) = \frac{3(1+v)}{2(1-v)}, \quad (13)$$

and the optimal competitive ratio is

$$\boxed{\text{CR}_{\text{cap}}^*(v) = \frac{27(1+v)}{2(1-v)^2} - \frac{1-v}{1+v} = \frac{(2v+1)(v+5)^2}{2(1-v)^2(1+v)}}. \quad (14)$$

This is also a lower bound for arbitrary deterministic alternating record strategies.

Upper bound. It suffices to minimize

$$K(r) = \frac{r^3}{(1-v)r - (1+v)}.$$

The derivative vanishes when

$$3r^2((1-v)r - (1+v)) - (1-v)r^3 = 0,$$

which reduces to

$$2(1-v)r = 3(1+v).$$

This gives (13). At this point

$$K^* = \frac{27(1+v)^2}{4(1-v)^3},$$

and substitution in (12) yields (14). $\square$

For the matching lower bound, an arbitrary alternating record sequence satisfies

$$B_i(v) = (1-v)S_i - (1+v)S_{i-1}.$$

Placing the target just beyond $B_{i-2}(v)$ yields

$$\text{CR} \geq \frac{1-v}{1+v} \left(2 \sup_i \frac{S_i}{(1-v)S_{i-2} - (1+v)S_{i-3}} - 1 \right). \quad (15)$$

The sharp sequence constant is

$$K_{\text{cap}}^*(v) = \min_{r > (1+v)/(1-v)} \frac{r^3}{(1-v)r - (1+v)} = \frac{27(1+v)^2}{4(1-v)^3},$$

identified by the critical characteristic equation

$$r^3 = K((1-v)r - (1+v)).$$

This matches the geometric upper bound. For $v = 0$, (14) gives $25/2$; as $v \rightarrow 1^-$,

$$\text{CR}_{\text{cap}}^*(v) \sim \frac{27}{(1-v)^2}.$$

5 Discovery with known distance and unknown speed

We now assume that $D = |d|$ is known but the sign σ and the speed v are unknown. By scaling, set $D = 1$. A uniform constant competitive ratio is not the appropriate objective as $v \rightarrow 1^-$. We seek a bound depending on the actual speed.

5.1 Doubly-exponential strategy and induced speed thresholds

Fix constants

$$A > 1, \quad b > 1,$$

and use alternating excursions

$$\boxed{a_i = A^{b^i}}. \tag{16}$$

This strategy is implementable without knowing v .

Instead of prescribing guesses independently, define the maximal speed handled by round i through the critical equality

$$1 + v_i t_i = a_i.$$

Using (1),

$$\boxed{v_i = \frac{a_i - 1}{a_i + 2S_{i-1}}, \quad \delta_i := 1 - v_i = \frac{1 + 2S_{i-1}}{a_i + 2S_{i-1}}}. \tag{17}$$

Because the sequence is doubly exponential,

$$S_i = a_i(1 + o(1)), \quad \frac{a_{i-1}}{a_i} = A^{-(b-1)b^{i-1}}. \tag{18}$$

Hence

$$\boxed{\delta_i = 1 - v_i \sim 2 \frac{a_{i-1}}{a_i} = 2A^{-(b-1)b^{i-1}}}. \tag{19}$$

In particular $v_i \rightarrow 1$, so every fixed $v < 1$ is eventually dominated by the induced guesses.

For a fixed actual speed define

$$i(v) := \max\{i : v_i < v\}. \tag{20}$$

Thus

$$v_{i(v)} < v \leq v_{i(v)+1}.$$

Round $i(v) + 1$ is speed-sufficient. Since the sign is unknown and excursions alternate, the adversary can put the target on the other side; round $i(v) + 2$ is then guaranteed to be both on the correct side and to have a still larger speed threshold. Thus the worst-sign discovery time is bounded by

$$T_{\text{disc}}(v) \leq t_{i(v)+2}. \quad (21)$$

Theorem 5.1 (Unknown-speed discovery upper bound). *For the strategy (16),*

$$\text{CR}_{\text{disc}}(v) = O\left((1-v)^{-\alpha(b)}\right), \quad \alpha(b) := \frac{b^3}{b-1} - 1, \quad (22)$$

as $v \rightarrow 1^-$. The exponent is minimized uniquely at $b = 3/2$, giving

$$\boxed{\text{CR}_{\text{disc}}(v) = O\left((1-v)^{-23/4}\right)}. \quad (23)$$

Proof. Let $i = i(v)$. Since $v_i < v$, we have

$$\delta := 1 - v < \delta_i.$$

Since $v \leq v_{i+1}$,

$$\delta \geq \delta_{i+1}.$$

On the interval of speeds with the same transition index i , the discovery ratio

$$\frac{t_{i+2}}{1/(1-v)} = \delta t_{i+2}$$

is increasing in δ . Therefore its worst value in that interval occurs as $\delta \uparrow \delta_i$.

Using (19) and $t_{i+2} \sim a_{i+2}$,

$$\begin{aligned} \text{CR}_{\text{disc}}(v) &\leq (1 + o(1))\delta_i a_{i+2} \\ &\sim 2 \frac{a_{i-1}}{a_i} a_{i+2}. \end{aligned}$$

Now

$$a_{i-1} = A^{b^{i-1}}, \quad a_i = A^{b^i}, \quad a_{i+2} = A^{b^{i+2}},$$

so

$$\delta_i a_{i+2} \asymp A^{[b^3 - (b-1)]b^{i-1}}.$$

Meanwhile

$$\delta_i^{-1} \asymp A^{(b-1)b^{i-1}}.$$

Therefore

$$\delta_i a_{i+2} = \Theta \left(\delta_i^{-\left(\frac{b^3}{b-1}-1\right)} \right),$$

which proves (22); replacing δ_i by any $\delta \in [\delta_{i+1}, \delta_i)$ changes only the interval-wise constant at the stated power exponent.

Finally,

$$\alpha'(b) = \frac{b^2(2b-3)}{(b-1)^2},$$

so the unique minimizer for $b > 1$ is $b = 3/2$. At this value

$$\alpha(3/2) = \frac{(3/2)^3}{1/2} - 1 = \frac{27}{4} - 1 = \frac{23}{4}.$$

□

Remark 5.2. The base A is fixed in advance and does not depend on the unknown actual speed. One may interpret A^{b^i} as generating increasingly aggressive implicit speed guesses. Formula (17) is the precise relationship between the spatial excursion and the speed guess it supports.

6 Capture with known distance and unknown speed

Use the same doubly-exponential alternating strategy and the same transition index $i(v)$. In the worst sign, discovery occurs by round $k = i(v) + 2$. From (10) with $D = 1$,

$$T_{\text{cap}}(v) \leq \frac{2S_{i(v)+2} - 1}{1 + v}.$$

Dividing by $1/(1-v)$ gives

$$\text{CR}_{\text{cap}}(v) \leq \frac{1-v}{1+v} (2S_{i(v)+2} - 1). \quad (24)$$

Theorem 6.1 (Unknown-speed capture upper bound). *For $a_i = A^{b^i}$,*

$$\text{CR}_{\text{cap}}(v) = O\left((1-v)^{-\alpha(b)}\right), \quad \alpha(b) = \frac{b^3}{b-1} - 1. \quad (25)$$

The exponent is minimized at $b = 3/2$, and therefore

$$\boxed{\text{CR}_{\text{cap}}(v) = O\left((1-v)^{-23/4}\right), \quad v \rightarrow 1^-} \quad (26)$$

Proof. The proof is identical at the exponent level to Theorem 5.1. On a fixed threshold interval the factor $(1 - v)/(1 + v)$ is at most $1 - v$, while $S_{i+2} \sim a_{i+2}$. Thus

$$\text{CR}_{\text{cap}}(v) \leq (1 + o(1))(1 - v)a_{i+2}.$$

The calculation in the preceding proof gives exponent $\alpha(b) = b^3/(b - 1) - 1$, minimized by $b = 3/2$ with value $23/4$. $\square$

7 Summary of the four regimes

The results can be summarized as follows.

Information and objective	Result	Status
D unknown, v known; discovery	Eq. (7)	tight upper/lower
D unknown, v known; capture	Eq. (14)	tight upper/lower
D known, sign and v unknown; discovery	$O((1 - v)^{-23/4})$	upper bound
D known, sign and v unknown; capture	$O((1 - v)^{-23/4})$	upper bound

Here “ D known” means that the magnitude $D = |d|$ is known but the sign of d is not.

8 A conjectured matching lower exponent for unknown speed

The exponent $23/4$ is rigorously optimal within the constant-exponent doubly-exponential family $a_i = A^{b^i}$. It is natural to ask whether it is also unavoidable for arbitrary deterministic strategies.

Normalize $D = 1$ and consider an arbitrary alternating record strategy. Define

$$\delta_i := 1 - v_i = \frac{1 + 2S_{i-1}}{a_i + 2S_{i-1}} = \frac{1 + 2S_{i-1}}{S_i + S_{i-1}}.$$

When $S_i \gg S_{i-1}$,

$$\delta_i \asymp \frac{S_{i-1}}{S_i}. \quad (27)$$

Take an actual speed just above v_i . Round i is insufficient. By choosing the adverse sign, the target can force the searcher to wait approximately two

further rounds. Both discovery and capture then have normalized cost of order

$$\delta_i S_{i+2}. \quad (28)$$

Introduce local logarithmic growth exponents

$$p_i := \frac{\log S_i}{\log S_{i-1}}. \quad (29)$$

Heuristically $S_i \approx S_{i-1}^{p_i}$. Equation (27) gives

$$\log \frac{1}{\delta_i} \approx (p_i - 1) \log S_{i-1},$$

while

$$\log S_{i+2} \approx p_i p_{i+1} p_{i+2} \log S_{i-1}.$$

Thus the local lower-bound exponent suggested by (28) is

$$\beta_i \approx \frac{p_i p_{i+1} p_{i+2}}{p_i - 1} - 1. \quad (30)$$

For the constant choice $p_i \equiv p$, this is

$$\beta(p) = \frac{p^3}{p - 1} - 1,$$

whose minimum is $23/4$ at $p = 3/2$.

Conjecture 8.1 (Three-step sequence lower bound). *Every universal deterministic alternating record strategy satisfies, along an infinite record subsequence,*

$$\limsup_{i \rightarrow \infty} \left(\frac{p_i p_{i+1} p_{i+2}}{p_i - 1} - 1 \right) \geq \frac{23}{4}. \quad (31)$$

If Conjecture 8.1 can be established with the exact threshold relation rather than the heuristic approximation (27), it would imply a matching lower exponent for both unknown-speed problems:

Conjecture 8.2 (Universal unknown-speed lower exponent). *For every deterministic universal landing-only strategy $\mathcal{A}$, there exists a sequence $v_k \rightarrow 1^-$ such that, for discovery and likewise for capture,*

$$\text{CR}_{\mathcal{A}}(v_k) = \Omega \left((1 - v_k)^{-23/4 + o(1)} \right).$$

Equivalently,

$$\limsup_{v \rightarrow 1^-} \frac{\log \text{CR}_{\mathcal{A}}(v)}{\log(1/(1 - v))} \geq \frac{23}{4}.$$

9 Discussion and open problems

The landing-only sensing restriction produces two qualitatively different regimes. When v is known, a fixed geometric expansion suffices and exact competitive ratios follow from a one-parameter optimization together with a matching sequence lower bound. Both discovery and capture grow as $\Theta((1 - v)^{-2})$ as $v \rightarrow 1^-$. When v is unknown, the searcher must hedge against speeds arbitrarily close to one. A fixed geometric expansion no longer suffices universally; doubly-exponential excursions generate speed thresholds tending to one and yield the exponent $23/4$.

Several questions remain. The most important is to prove or refute the conjectured $23/4$ lower exponent for arbitrary unknown-speed strategies. A second question is whether a nonstationary doubly-exponential scheme, with a varying local exponent rather than $b \equiv 3/2$, can improve lower-order factors while preserving the optimal exponent. Randomization is another natural direction: one may randomize the initial side, the phase of the excursion sequence, or the speed-threshold schedule. Finally, positive landing times or explicit costs for changing direction would connect the present model to turn-cost search and could change both the optimal expansion law and the benchmark.

Acknowledgment of proof status

The known-speed discovery and capture formulas are presented as tight results, with their lower bounds reduced to the corresponding sharp sequence lemmas/critical recurrences. The unknown-speed $23/4$ results are upper bounds for the stated doubly-exponential strategy. The proposed matching $23/4$ lower bound for arbitrary strategies is explicitly conjectural and is not used in proving the upper bounds.

References

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